LLM FINE-TUNING PIPELINE | PROJECT 1

Domain-Adapted Llama 3 8B

Software engineering Q&A with QLoRA

30.42% LOWER HELD-OUT ANSWER PERPLEXITY

Base 7.9560 -> Tuned 5.5359 | 100 paired examples | 18,798 answer tokens

What this project demonstrates

An end-to-end adaptation workflow that converts a large public preference corpus into a controlled supervised fine-tuning experiment, trains a parameter-efficient adapter on an A100, and compares the adapted and unadapted model under identical inference conditions.

50,500
real-world Q&A examples cached

0.26%
of parameters trained with LoRA

2:44:33
single-A100 training runtime

PORTFOLIO SIGNAL

The result is evidence that a small trainable adapter can materially shift an 8B model toward the target answer distribution while preserving a frozen quantized backbone. It demonstrates data engineering, efficient training, experimental control, statistical evaluation, and deployment-oriented artifact design.

From public Q&A to controlled SFT data

A deterministic pipeline selects stronger Stack Overflow answers, normalizes text, enforces quality gates, and isolates a fixed holdout before training.

SOURCE
10.8M-row Stack Exchange preference corpus

→

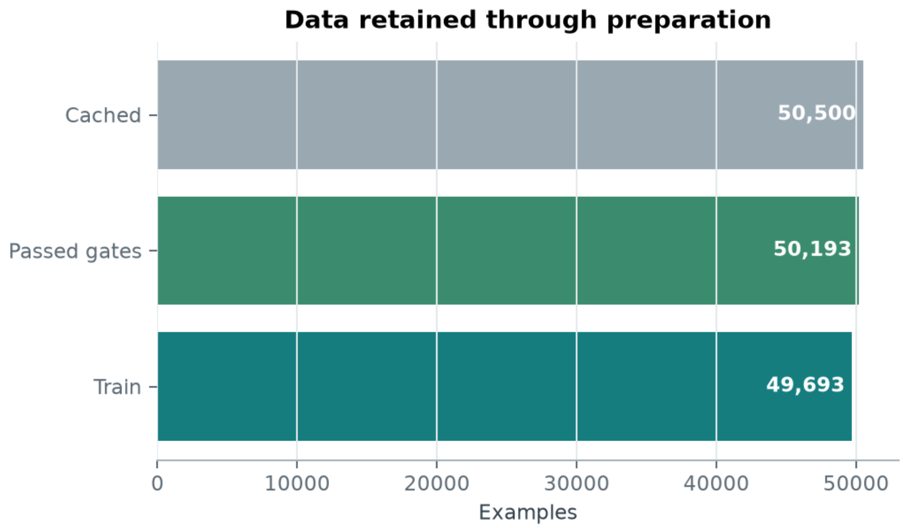
STREAM
Stackoverflow.com subset, seed 42 shuffle

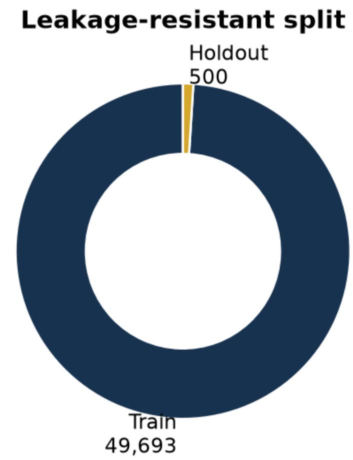
→

SELECT
pm_score >= 2; accepted answer prioritized

→

NORMALIZE
Strip HTML; retain metadata and CC BY-SA 4.0





Quality gates

Required fields
instruction and output must be non-empty

Length control
combined text <= 8,000 characters

Deduplication
case-normalized instruction/input/output key

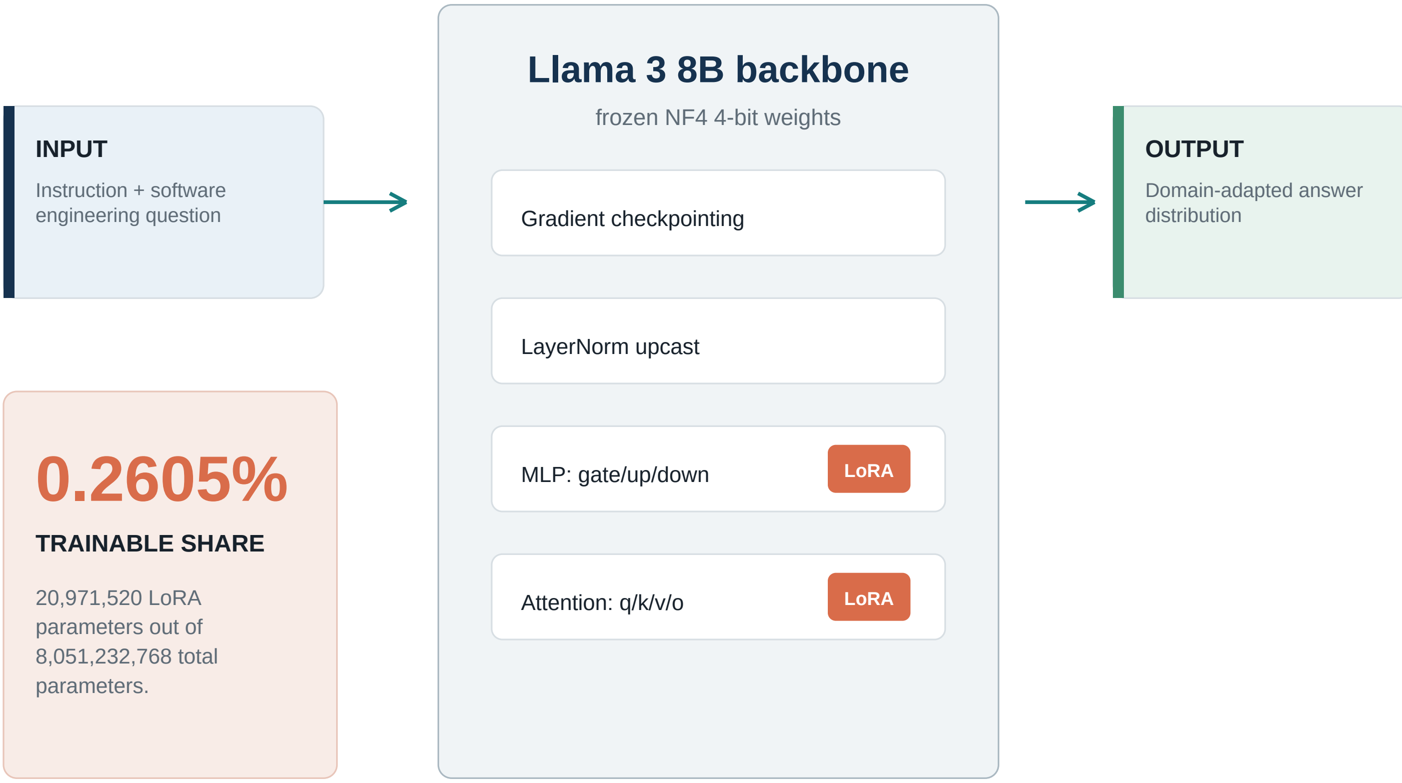
Outcome
307 rejected (0.61%); 50,193 retained

WHY THE SPLIT MATTERS

The 500-example holdout is removed before LLaMA-Factory registration and training. This prevents direct train/eval leakage. The final reported comparison samples 100 examples from that fixed holdout with seed 42; the remaining 400 are available for the final full-scale evaluation.

Efficient adaptation, not full retraining

QLoRA keeps the Llama 3 8B backbone frozen in 4-bit form while training low-rank updates on every linear projection.



Training configuration

BASE MODEL
Meta-Llama-3-8B-Instruct

SEQUENCE
2,048 tokens; packing enabled

METHOD
LoRA on all linear targets

BATCH
2 x 8 accumulation = 16 effective

QUANTIZATION
4-bit NF4 + double quantization

SCHEDULE
1 epoch; cosine; LR 2e-4

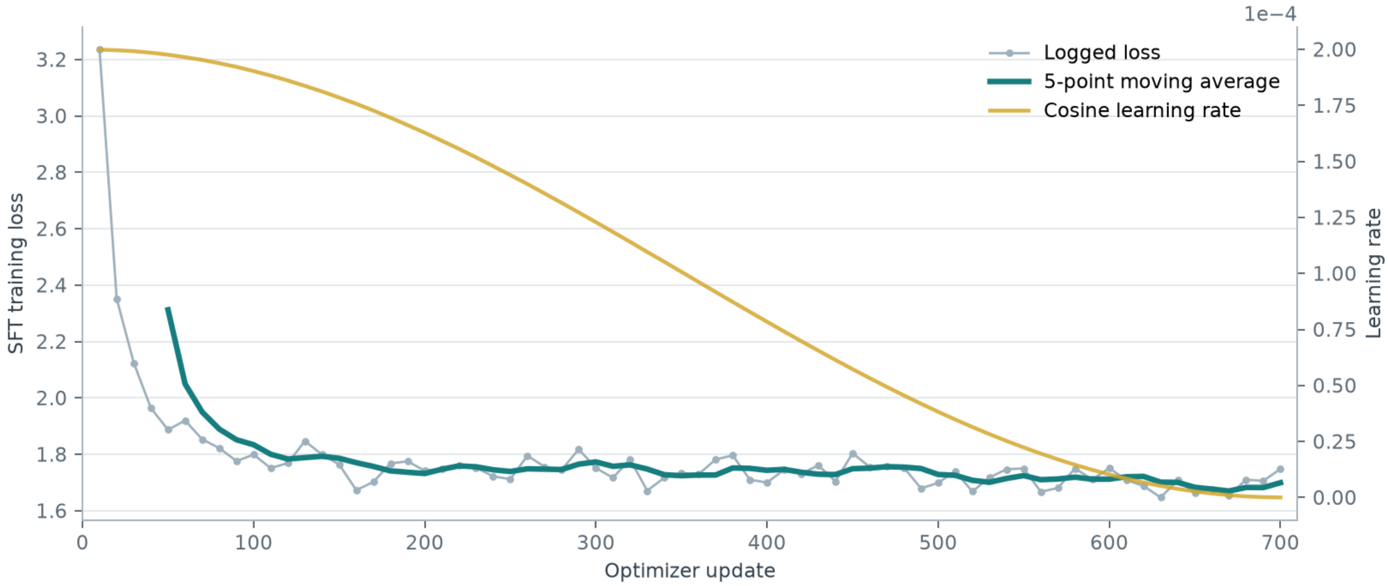
PRECISION
bfloat16 compute; LayerNorm FP32

CHECKPOINTS
every 50 steps; retain 2

Engineering meaning: the adapter is portable and inexpensive to store, while disabling it recovers the exact quantized baseline for a controlled A/B comparison.

Stable one-epoch adaptation on a single A100

The curve shows rapid early domain adaptation followed by a stable plateau under cosine decay.



701	**2:44:33**	**1.7811**	**1.135**
optimizer updates	wall-clock runtime	reported train loss	packed sequences/sec

OBSERVED DYNAMICS

Mean loss fell from 2.0732 across the first 10 logs to 1.6911 across the final 10, an 18.43% reduction. Gradient norms stabilized after the initial updates.

WHAT THE CURVE CANNOT PROVE

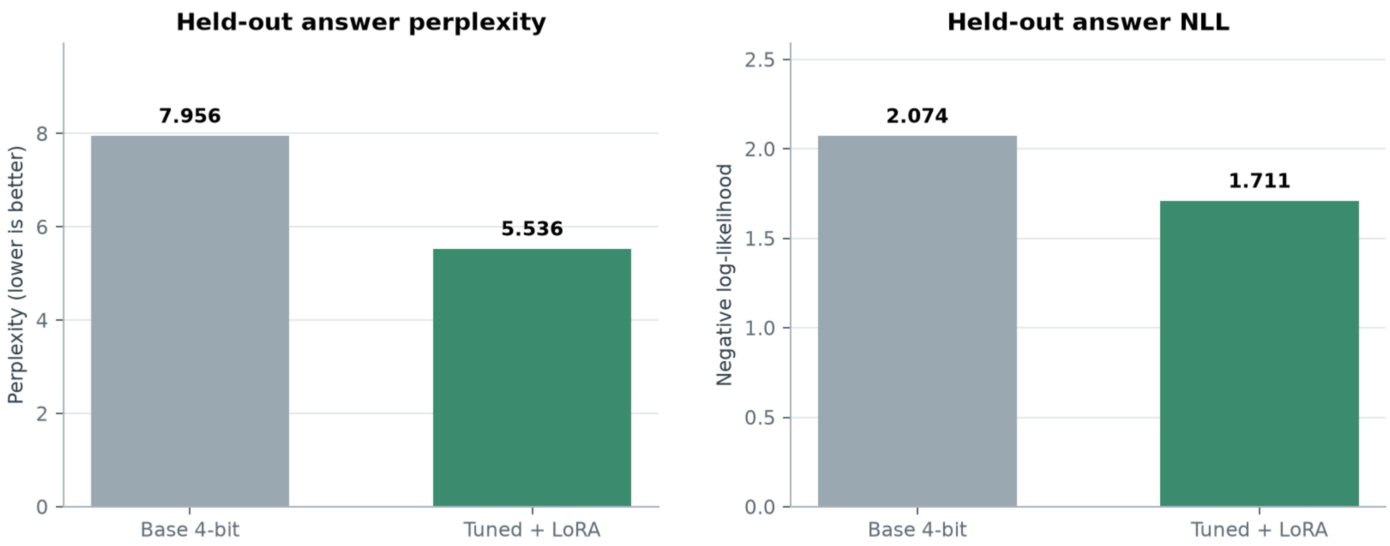
Training loss alone does not establish better answers. No in-training eval loss was recorded, so the claim depends on the separate frozen holdout evaluation shown next. This separation is deliberate and avoids tuning to the test metric.

A controlled paired comparison

The experiment changes one factor only: whether the trained LoRA adapter is active.

The adapter materially improves held-out fit

All displayed values come from the notebook's saved evaluation output.



30.42%

PERPLEXITY REDUCTION

7.9560 base -> 5.5359 tuned

100%

PAIRED EXAMPLE WIN RATE

100 examples; 18,798 answer tokens

CONFIDENCE CHECK

The 95% bootstrap interval for unweighted mean paired NLL improvement is [0.475145, 0.590001]. Because the entire interval is above zero, the direction of improvement is consistent across resampled evaluation sets.

DEFENSIBLE CLAIM

On a fixed 100-example holdout subset, QLoRA tuning reduced answer perplexity by 30.42% versus the same 4-bit base model.

DO NOT OVERCLAIM

Perplexity is not factual accuracy or production quality. The result shows stronger fit to held-out reference answers; human and execution-based evaluation remain necessary.

06 / TAKEAWAYS

What the project proves - and what comes next

A credible engineering report separates measured evidence from the next validation steps.

PROVEN IN THIS RUN

- Reproducible ingestion and atomic caching of 50,500 real-world records.
- Quality gates, deterministic split, and LLaMA-Factory registration.
- Single-A100 QLoRA training with 0.2605% trainable parameters.
- Controlled adapter-on versus adapter-off evaluation on held-out answers.
- 30.42% lower perplexity with a paired bootstrap interval above zero.

LIMITATIONS

- Final evaluation covers 100 of the 500 held-out examples.
- Reference likelihood does not directly score correctness, safety, or usefulness.
- Community votes and acceptance are imperfect proxies for answer quality.
- No independent contamination audit or executable-code benchmark was run.
- The adapter and evaluation artifacts live in ephemeral Colab storage unless exported.

NEXT EXPERIMENTS

- Run all 500 holdout examples and stratify by topic and answer length.
- Add human pairwise review for correctness, relevance, and code validity.
- Execute code-bearing answers in sandboxes and record pass rates.
- Benchmark latency, memory, and throughput before and after adapter merge.
- Publish the adapter, model card, data statement, and reproducibility manifest.

Evidence and source links

Project notebook	01_llm_fine_tuning_pipeline.ipynb
Stack Exchange preference dataset	https://huggingface.co/datasets/HuggingFaceH4/stack-exchange-preferences
Meta Llama 3 8B Instruct model card	https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct
LLaMA-Factory	https://github.com/hiyouga/LlamaFactory
QLoRA paper	https://arxiv.org/abs/2305.14314
PEFT adapter controls	https://huggingface.co/docs/peft/main/package_reference/peft_model
Perplexity definition	https://huggingface.co/docs/transformers/v4.33.0/en/perplexity

REPORTING RECOMMENDATION

Use the 30.42% figure with its metric and sample size: 'reduced held-out answer perplexity by 30.42% on a 100-example paired evaluation.' Replace it after the full 500-example run rather than presenting perplexity as a generic quality score.